

Simplifying Matching Methods for Causal Inference

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(Talk at MIT, Political Methodology Series, 3/16/2015)

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3 Problems, 3 Solutions

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 - ↪ “The Balance-Sample Size Frontier in Matching Methods for Causal Inference” (Gary King, Christopher Lucas and Richard Nielsen)

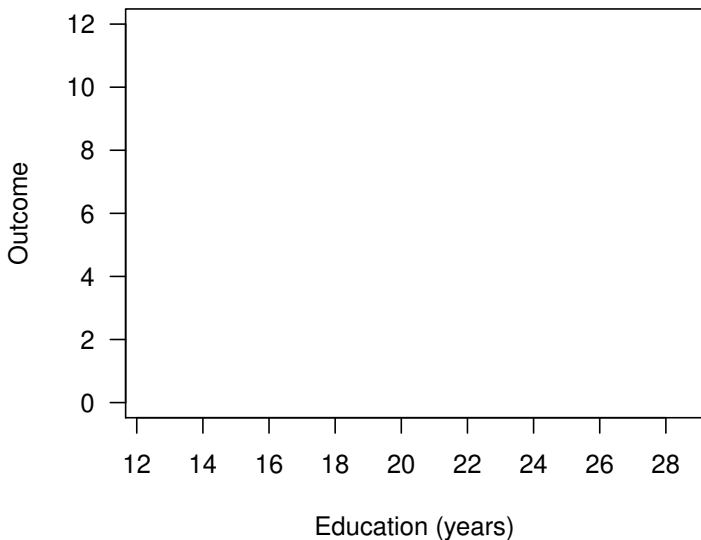
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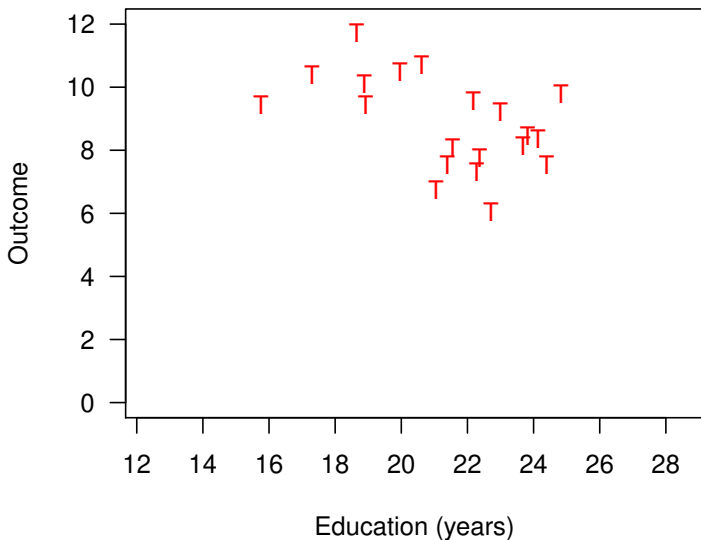
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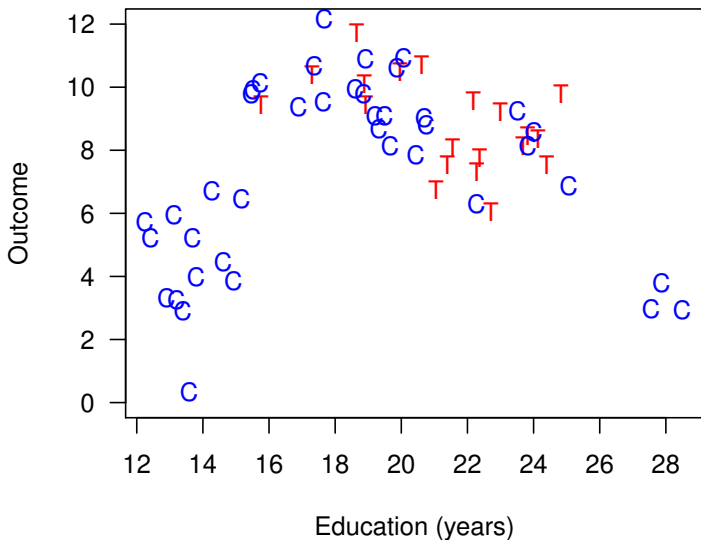
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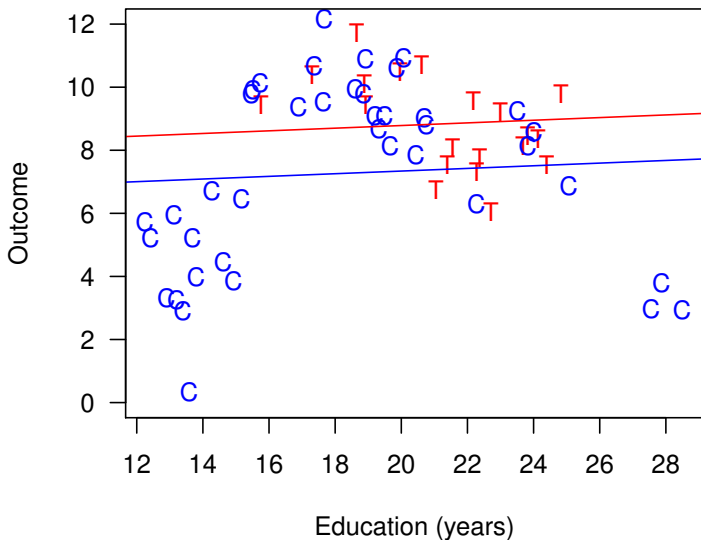
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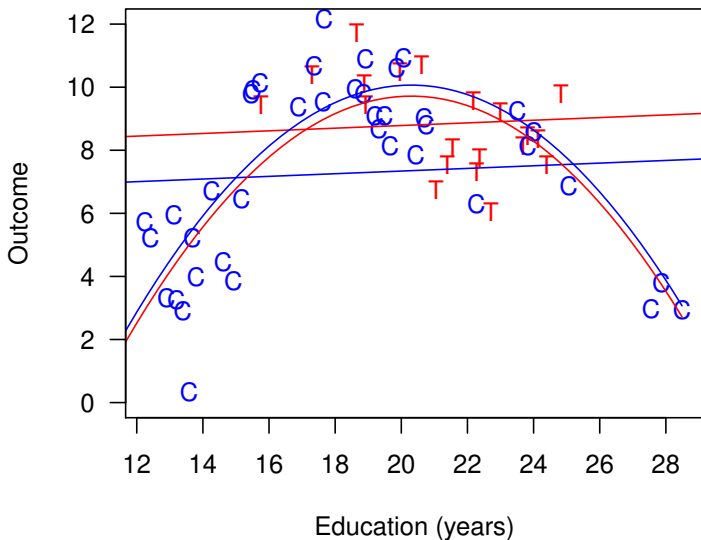
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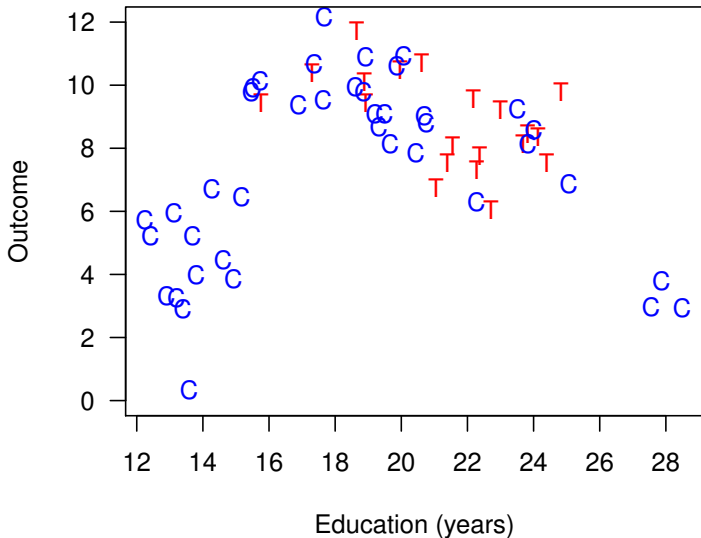
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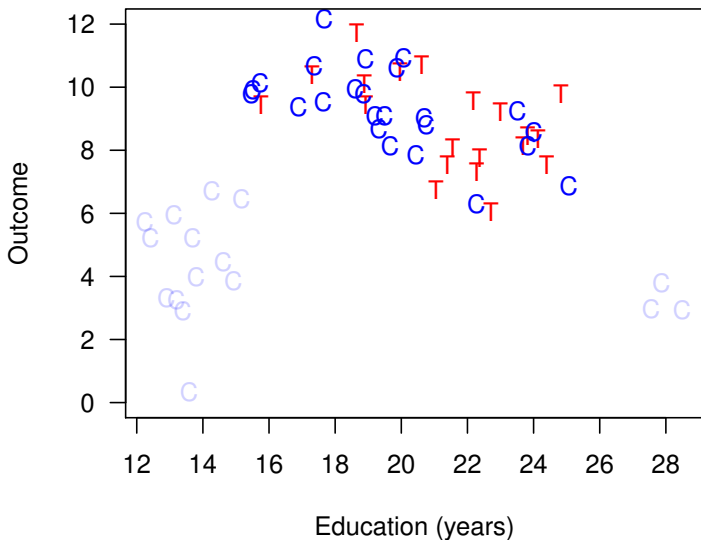
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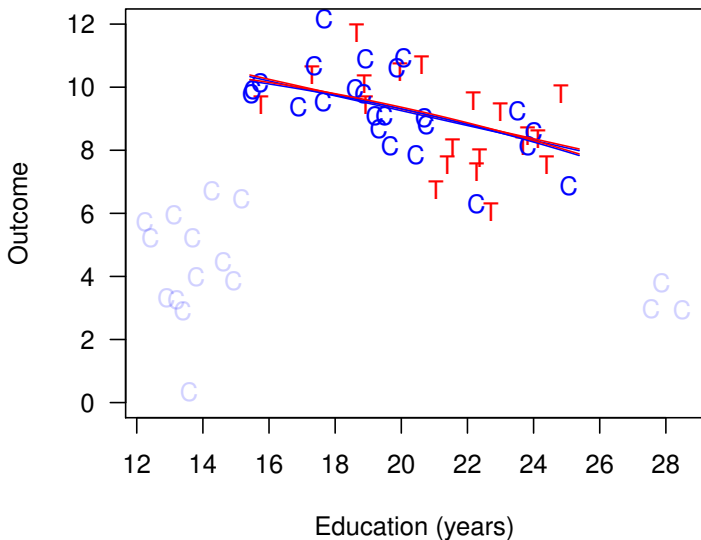
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- Big convenience: Follow preprocessing with whatever statistical method you'd have used without matching

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- Easy extensions for: multi-level, continuous, & mismeasured treatments; A too wide, n too small

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- Types of experiments:
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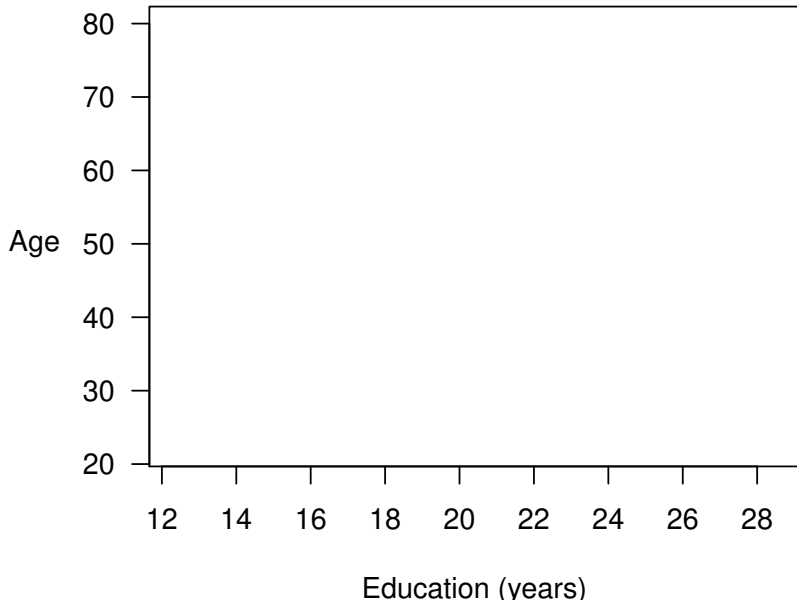
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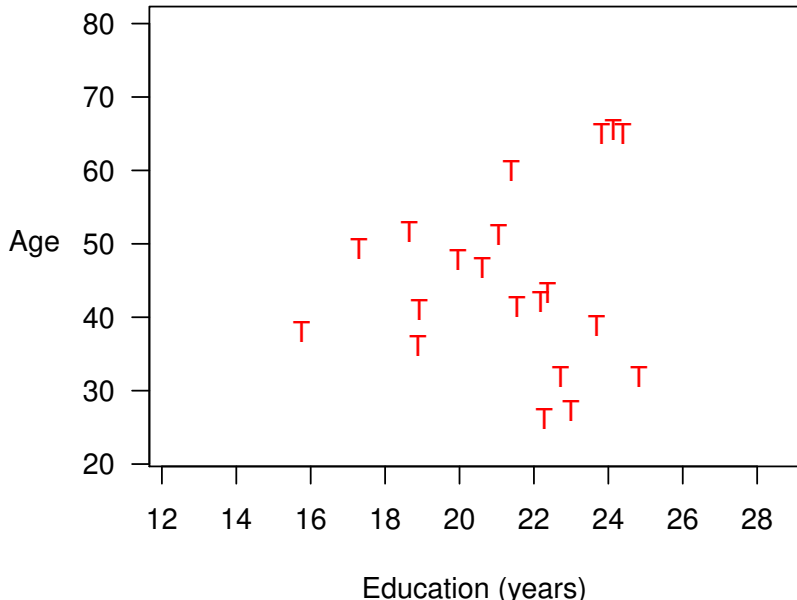
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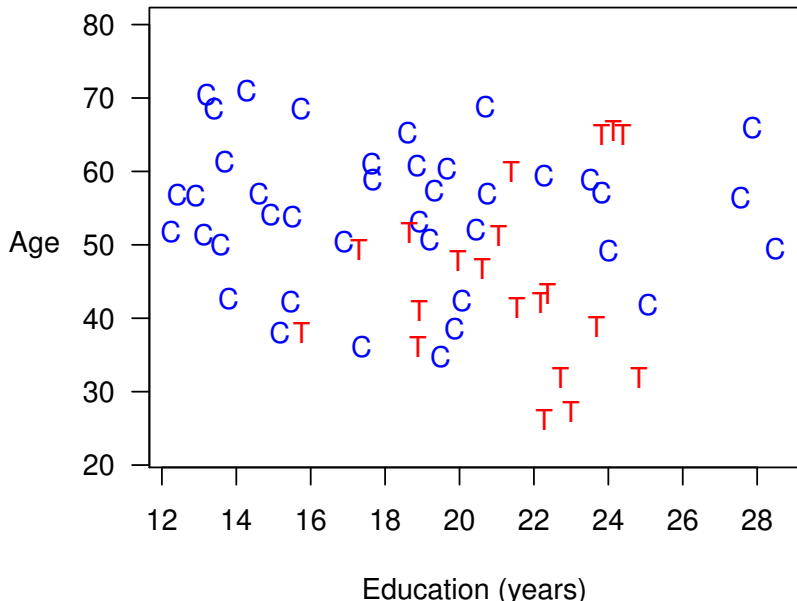
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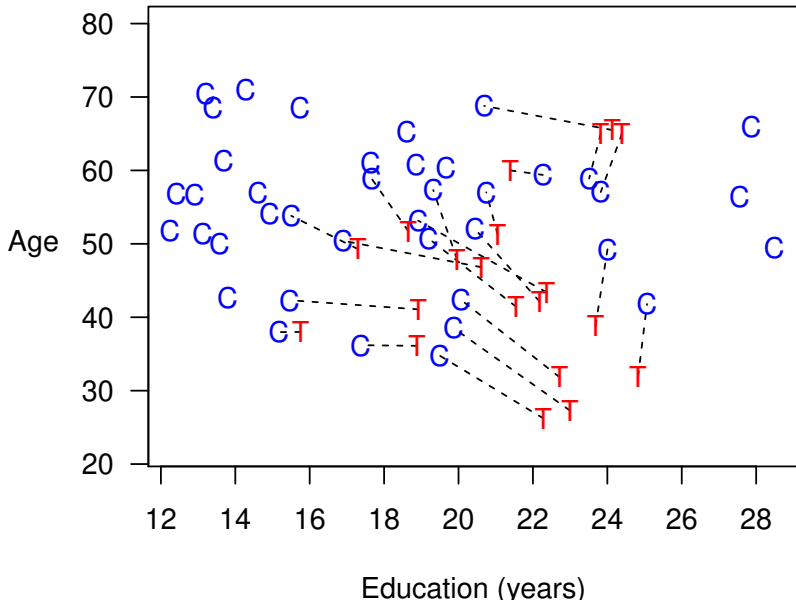
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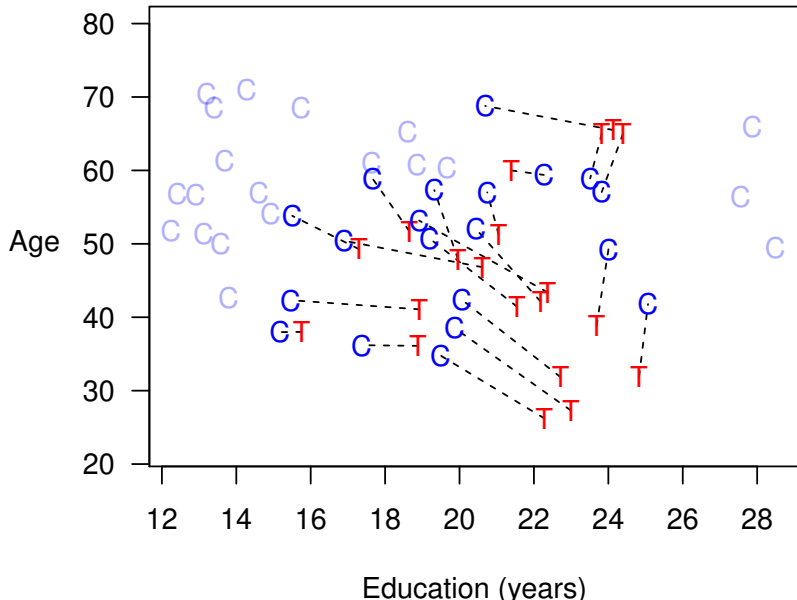
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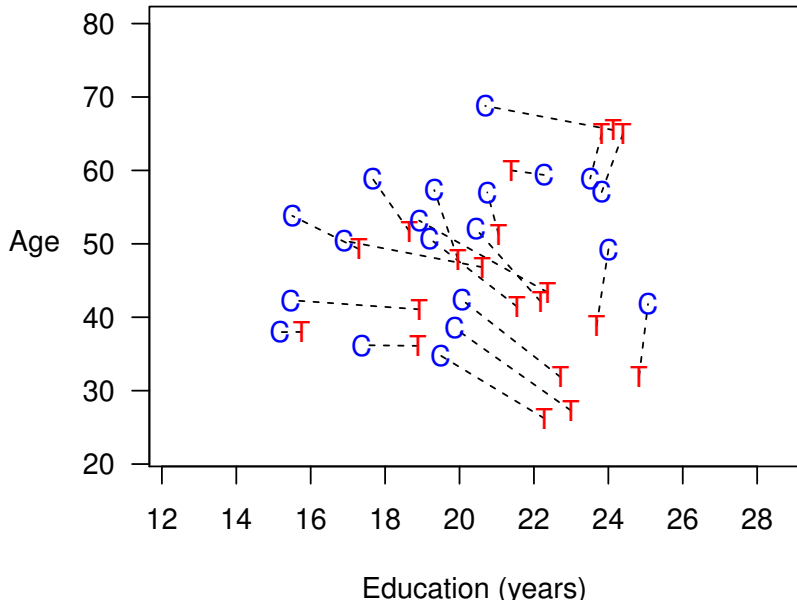
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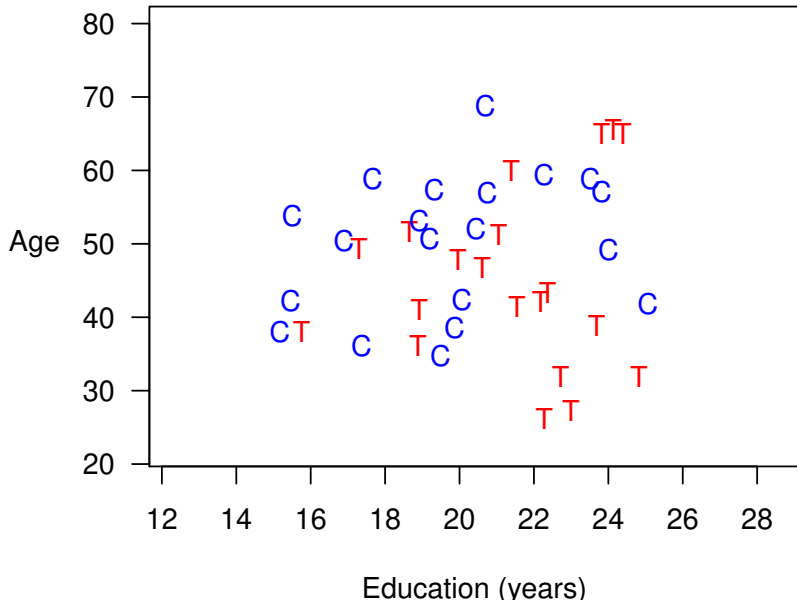
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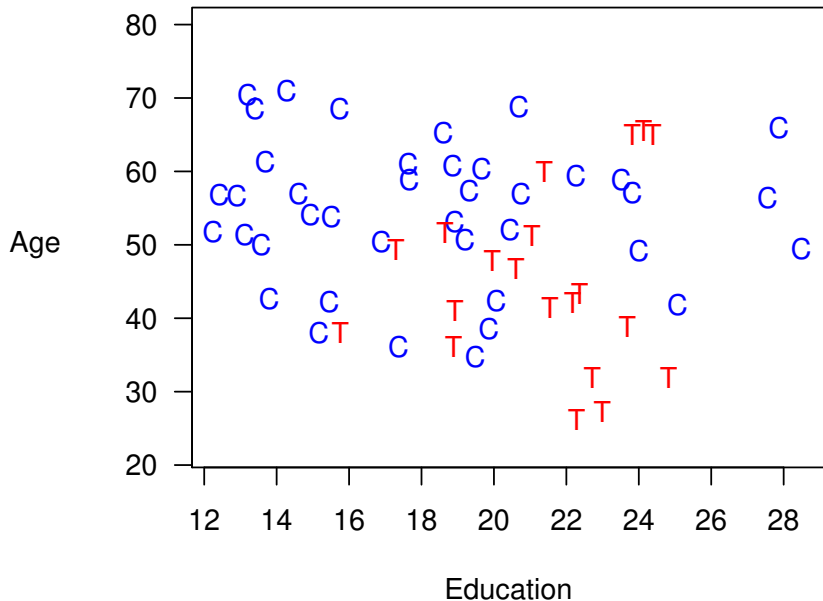
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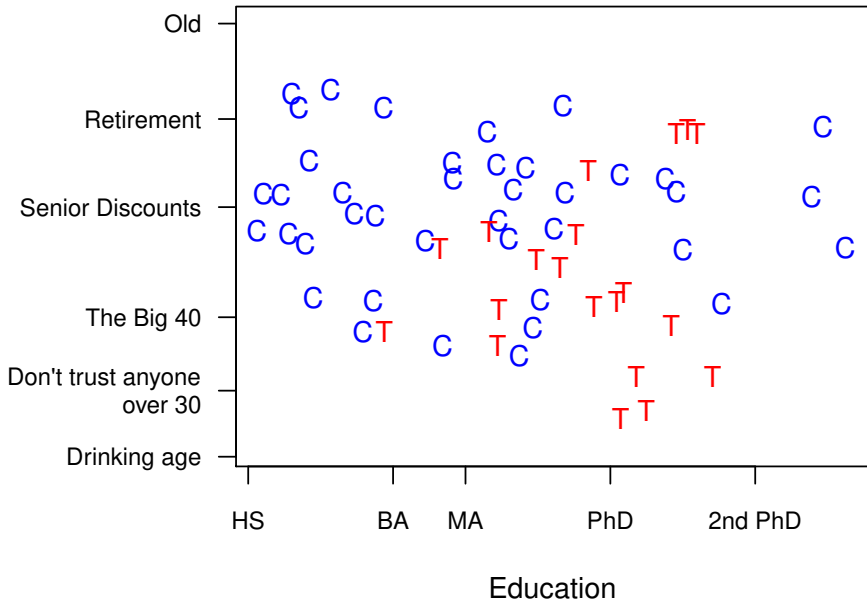
- Easier, but still iterative

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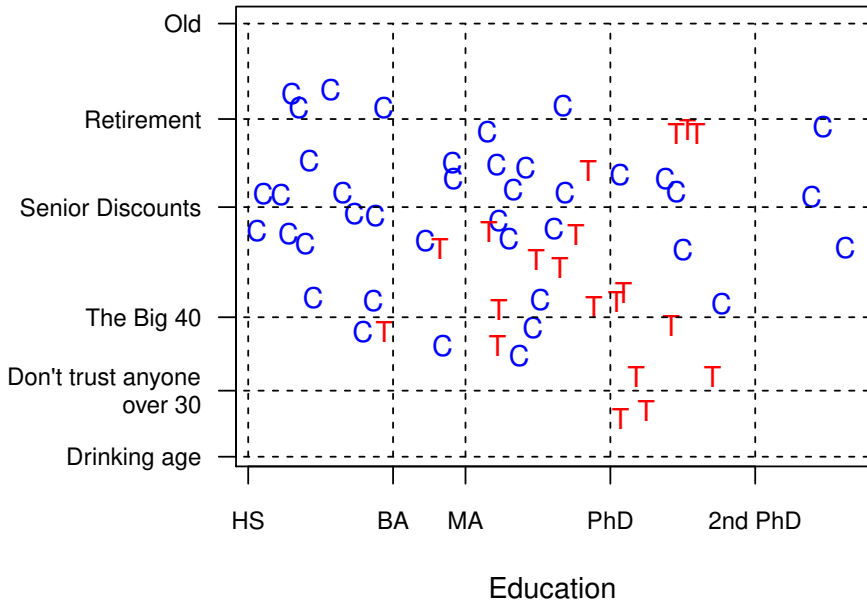
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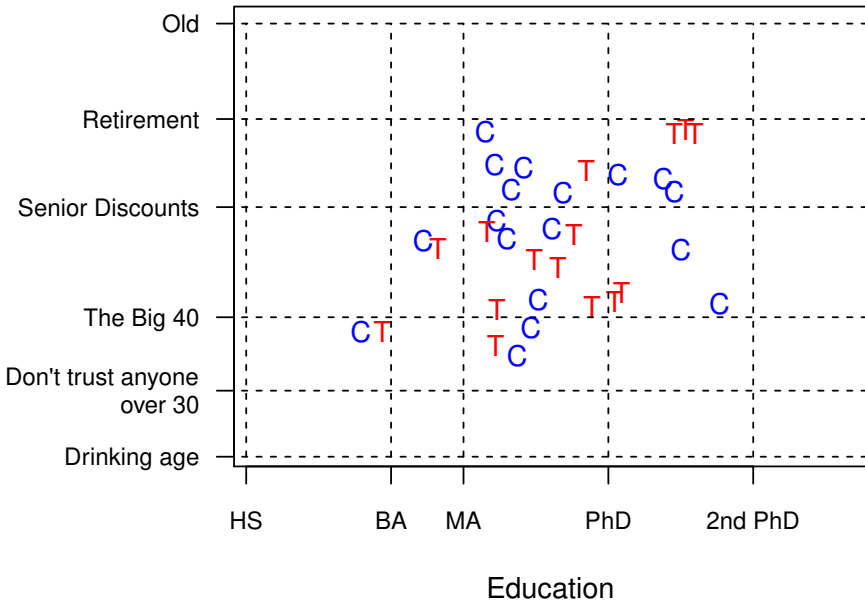
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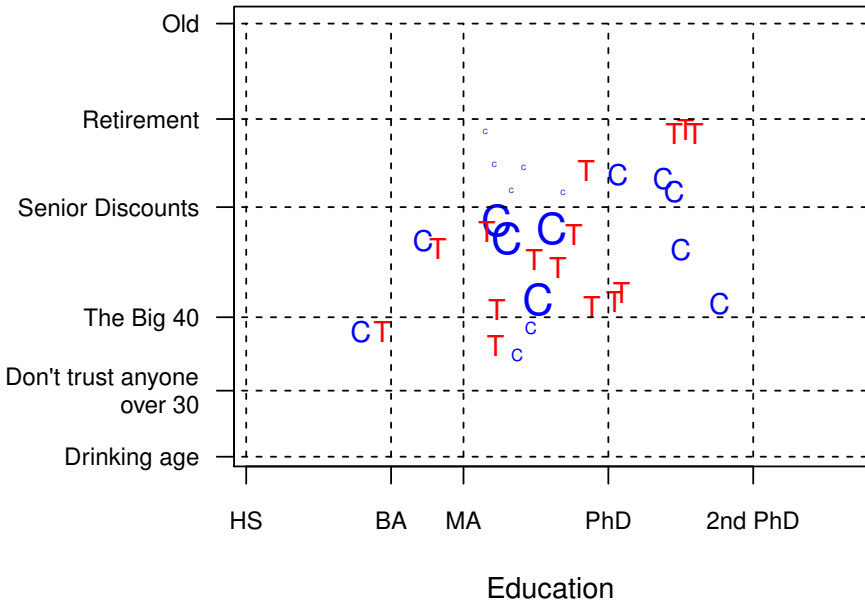
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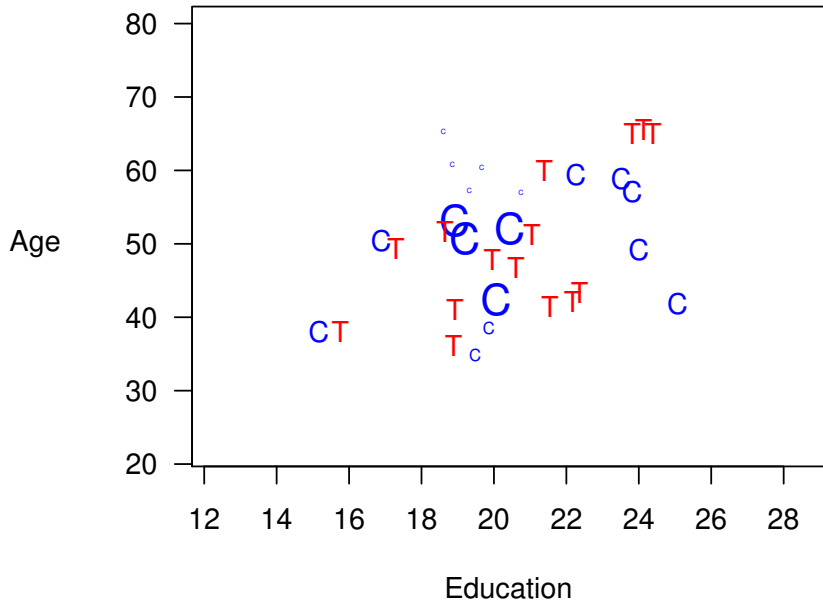
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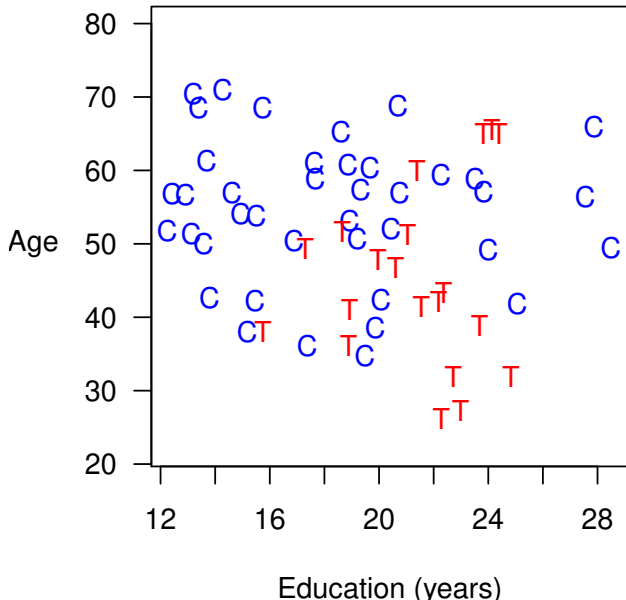
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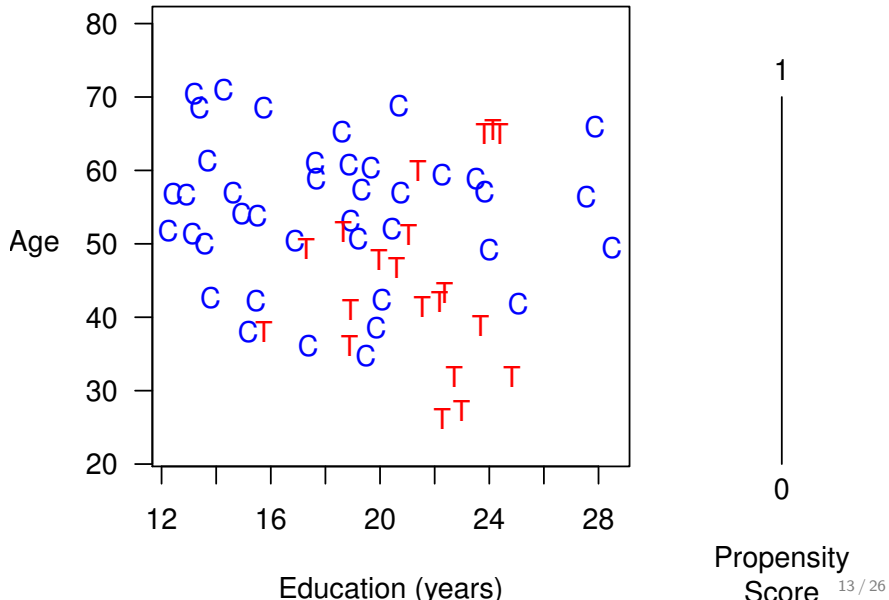
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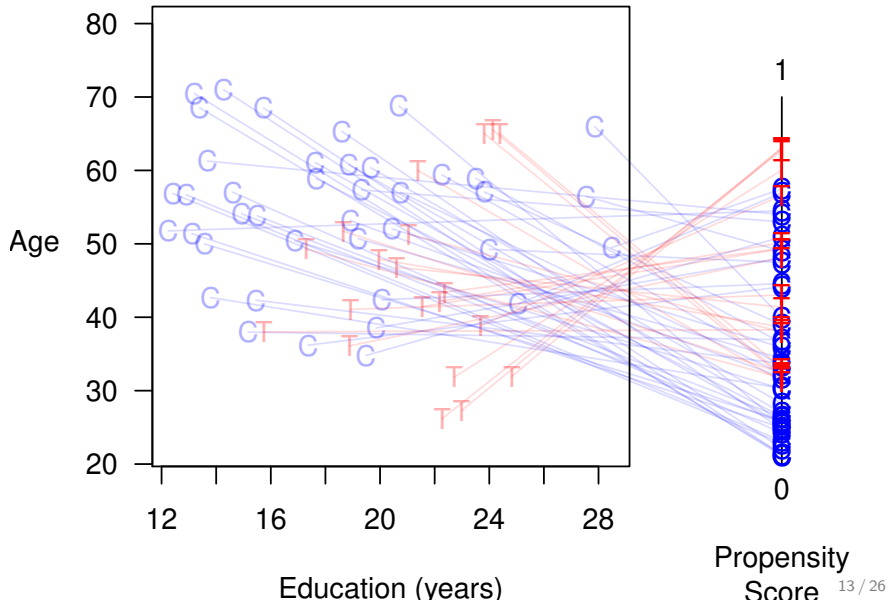
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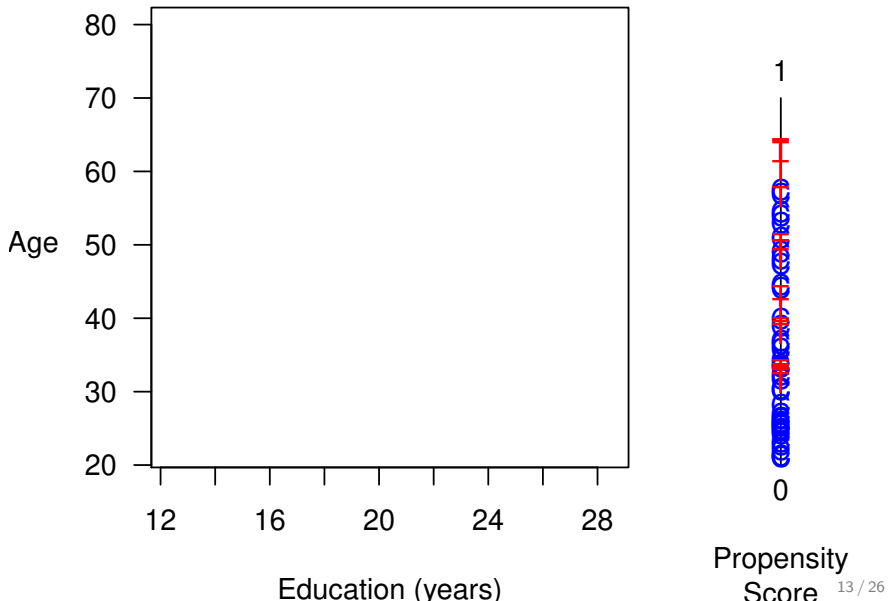
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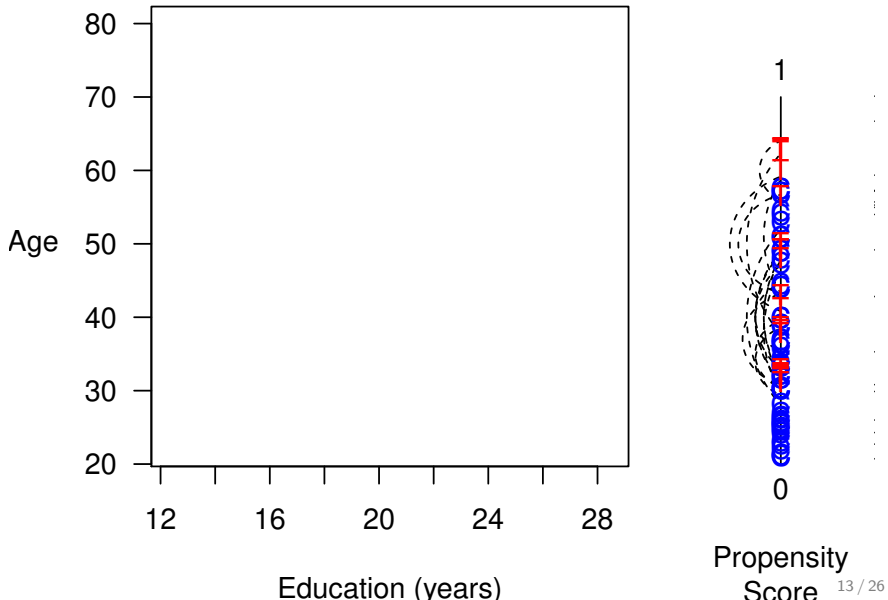
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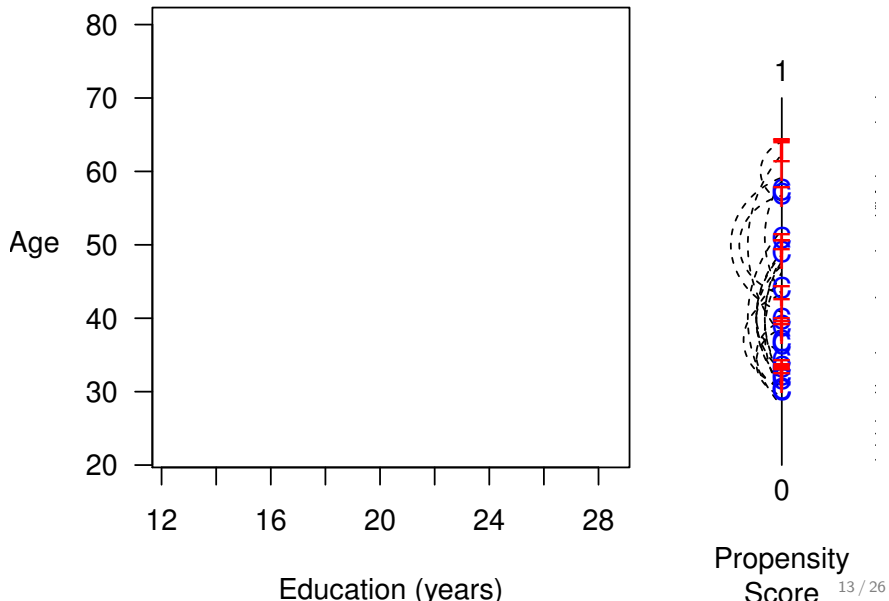
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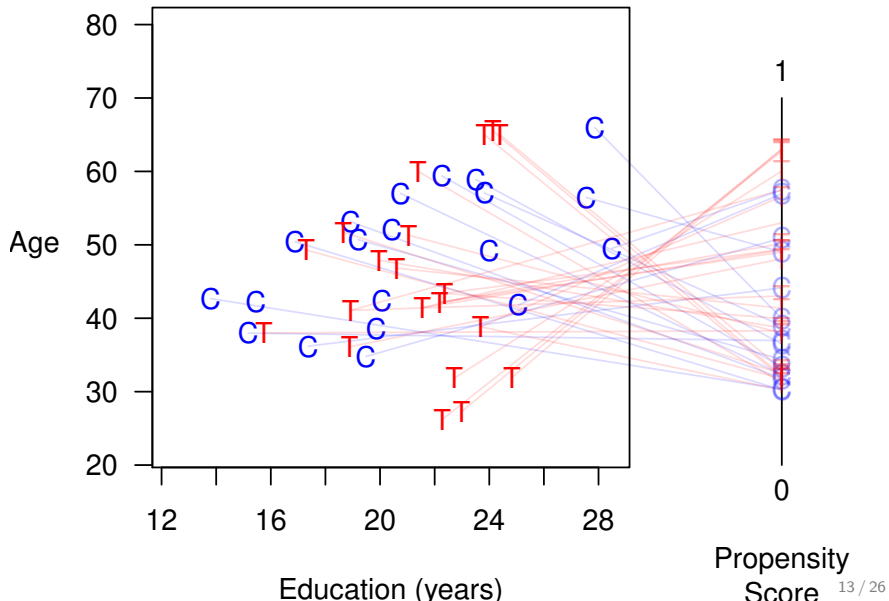
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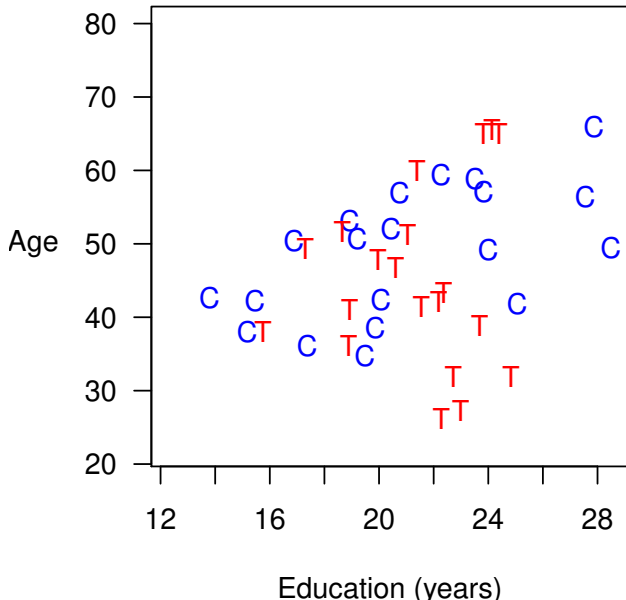
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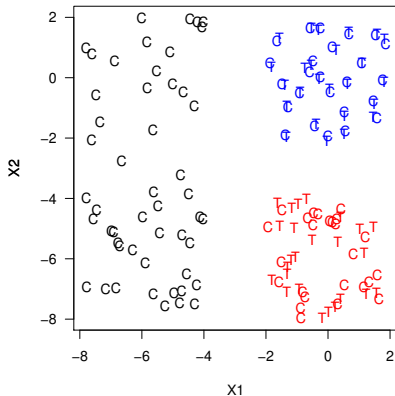
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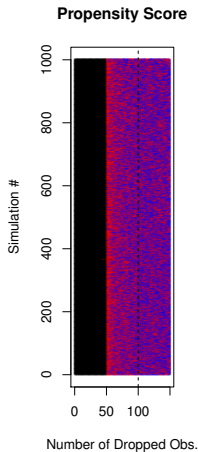
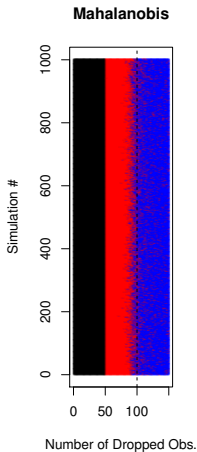
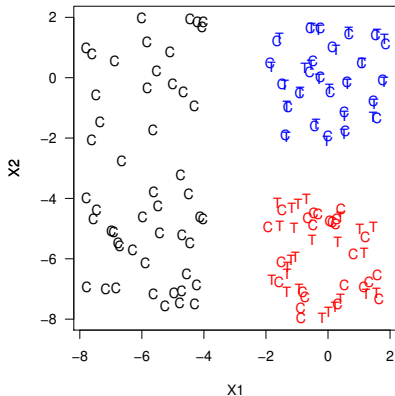
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 - The Reality: The PSM Paradox is bigger with more covariates

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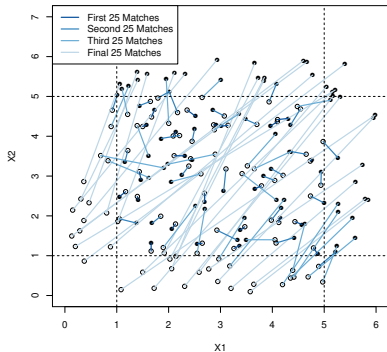


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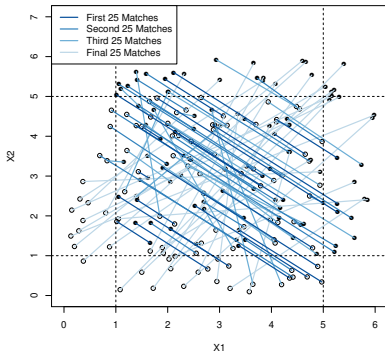


What Does PSM Match?

MDM Matches



PSM Matches

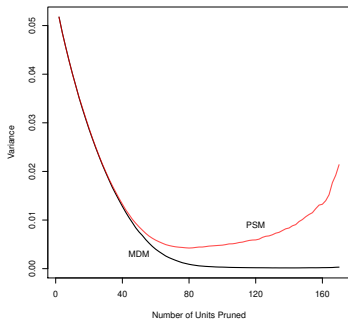


Controls: $X_1, X_2 \sim \text{Uniform}(0,5)$

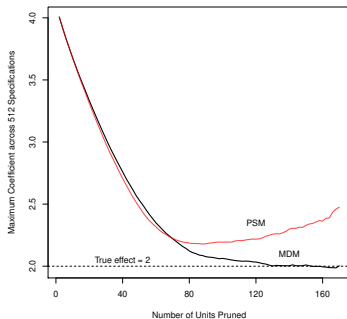
Treateds: $X_1, X_2 \sim \text{Uniform}(1,6)$

PSM Increases Model Dependence & Bias

Model Dependence



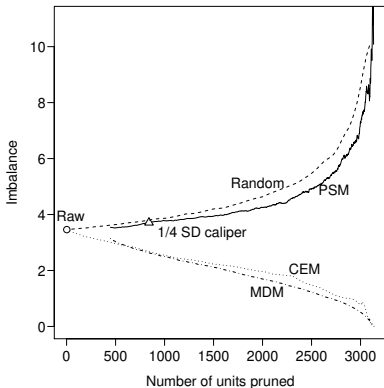
Bias



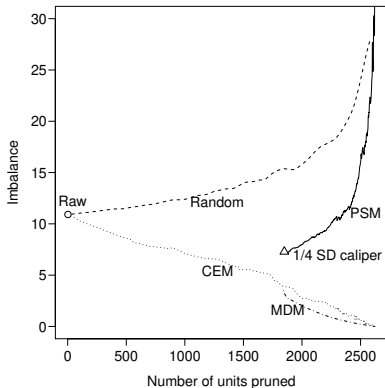
$$Y_i = 2T_i + X_{1i} + X_{2i} + \epsilon_i$$
$$\epsilon_i \sim N(0, 1)$$

The Propensity Score Paradox

Finkle et al. (2012)



Nielsen et al. (2011)



The Matching Frontier

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- Bias-Variance trade off $\rightsquigarrow$ Imbalance- n Trade Off

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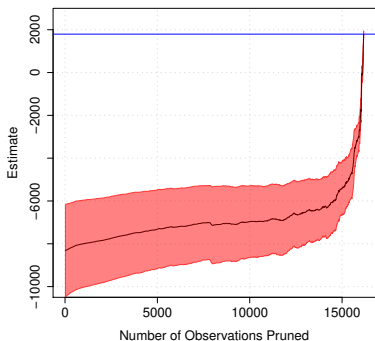
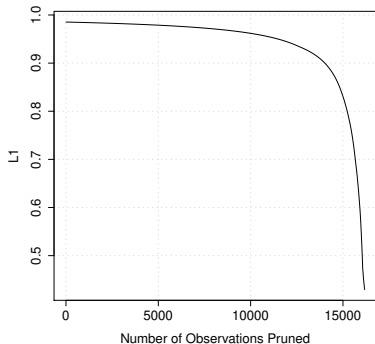
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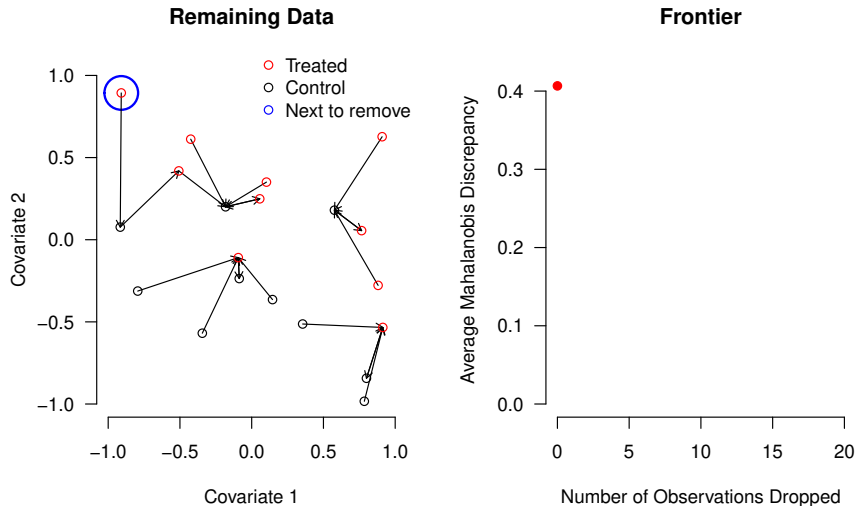
Job Training Data: Frontier and Causal Estimates



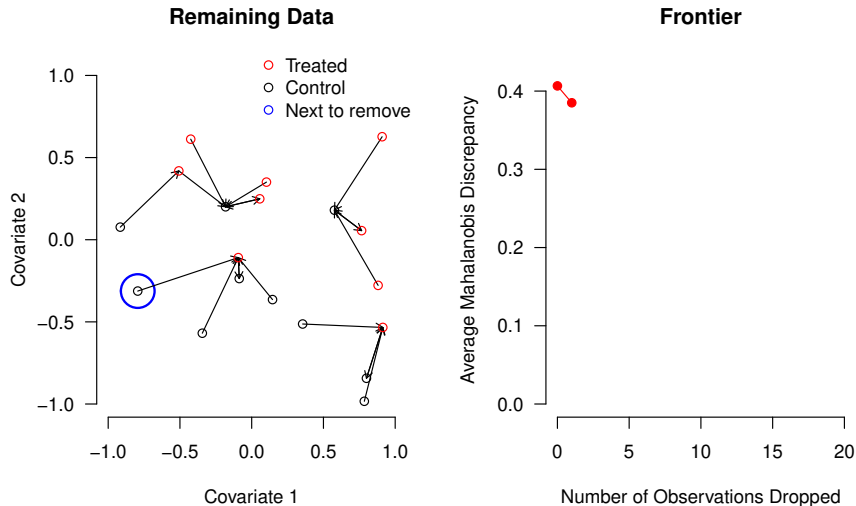
- 185 Ts; pruning most 16,252 Cs won't increase variance much
- Huge bias-variance trade-off after pruning most Cs
- Estimates converge to experiment after removing bias
- No mysteries: basis of inference clearly revealed

Constructing the FSATT Mahalanobis Frontier

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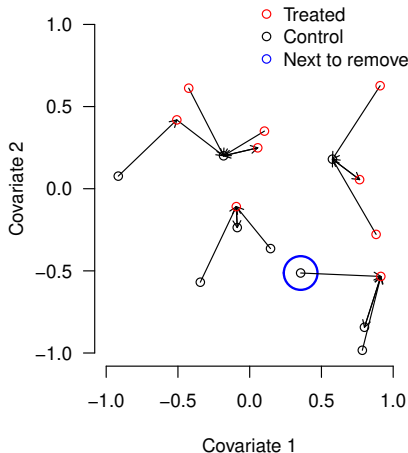


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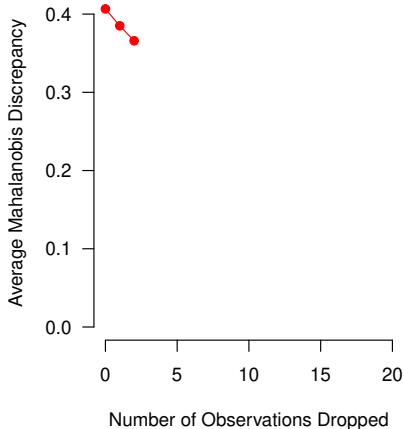


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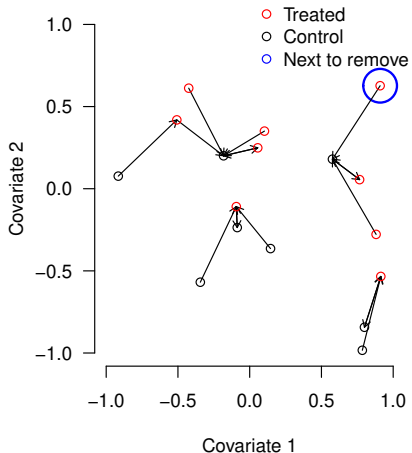


Frontier

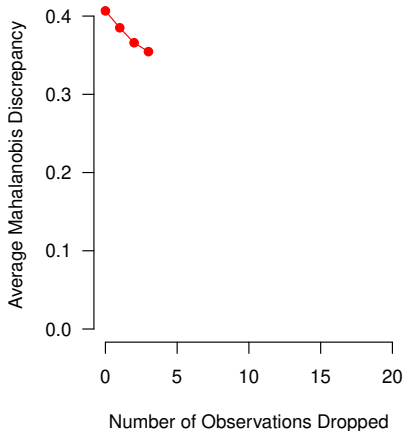


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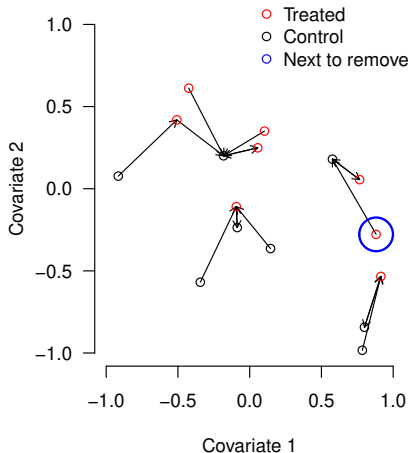


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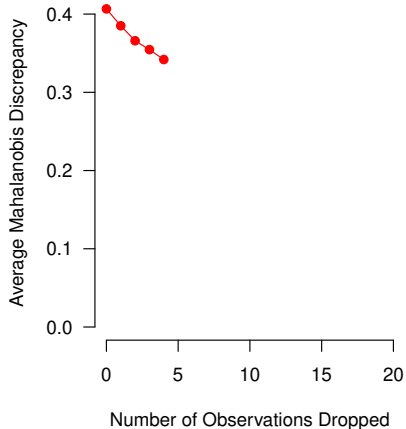


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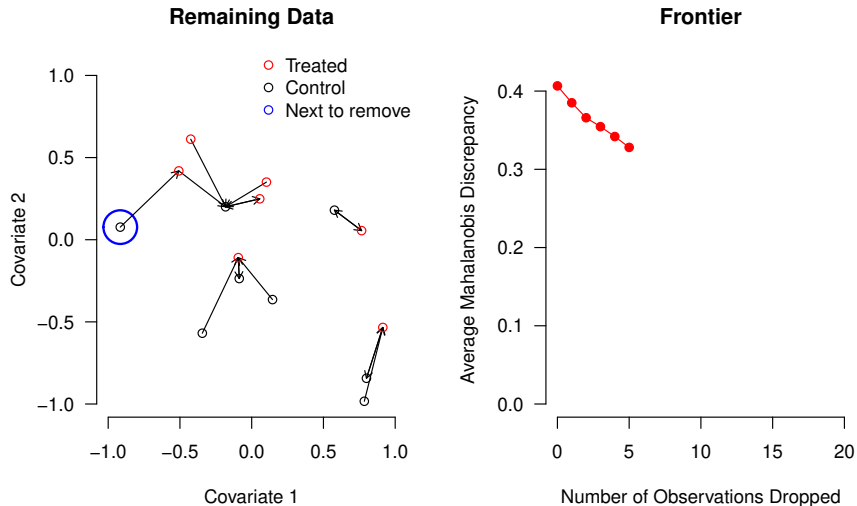
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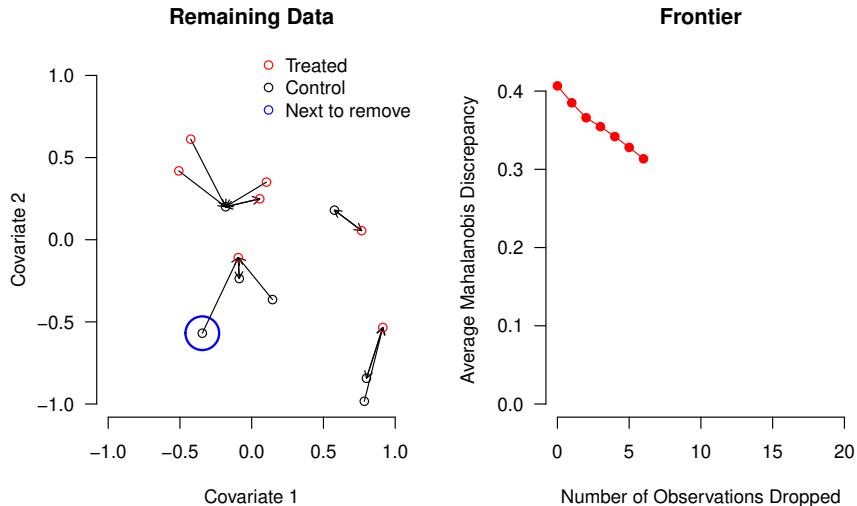
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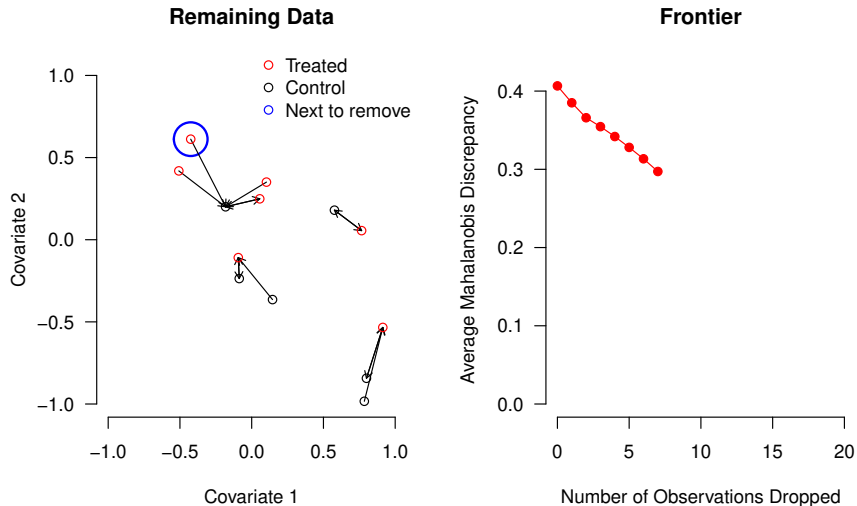
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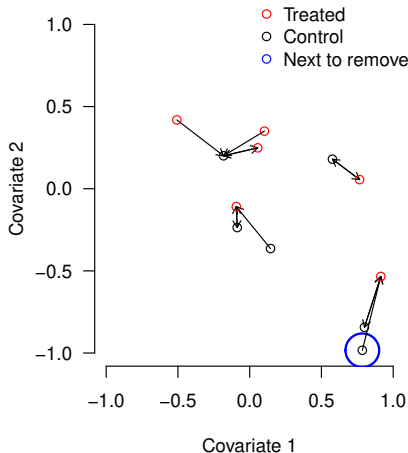


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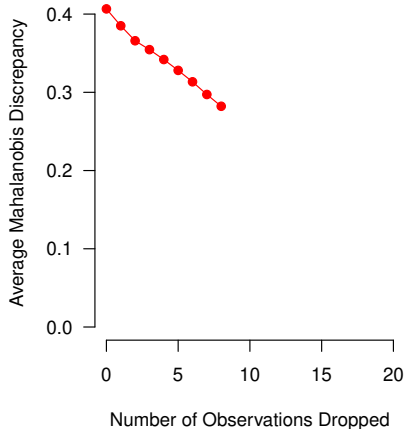


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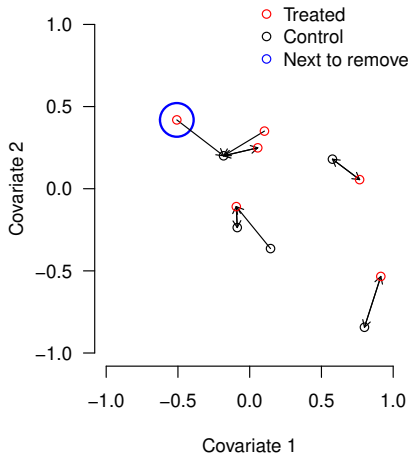


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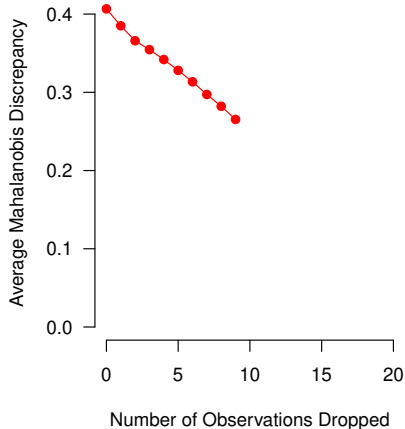


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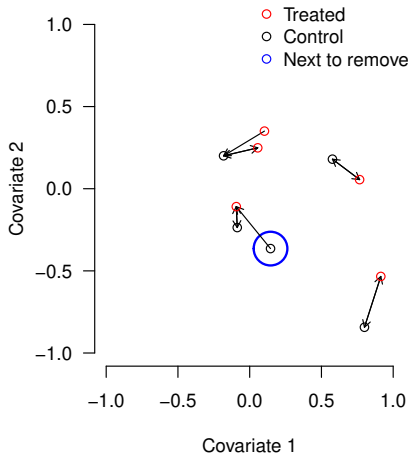


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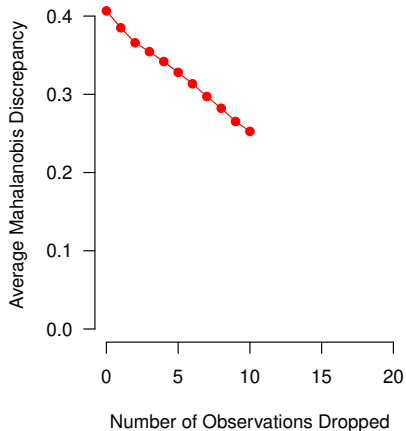


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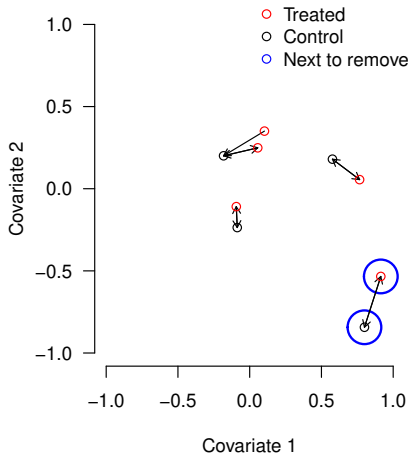


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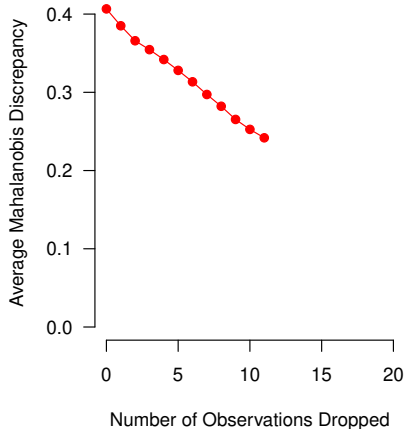


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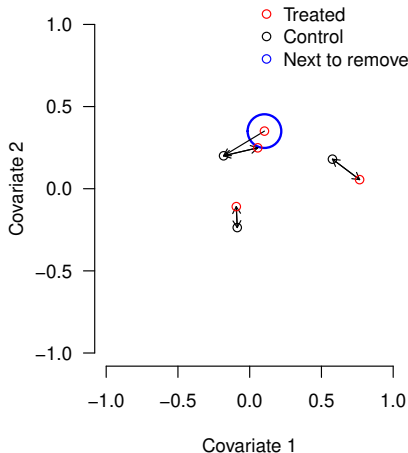


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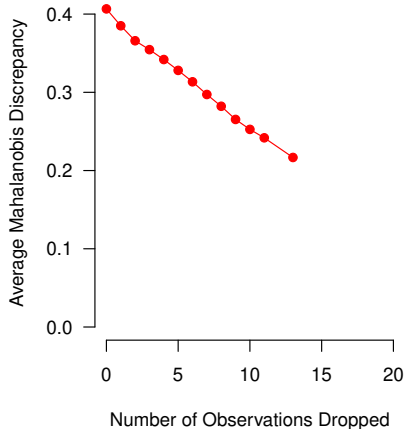


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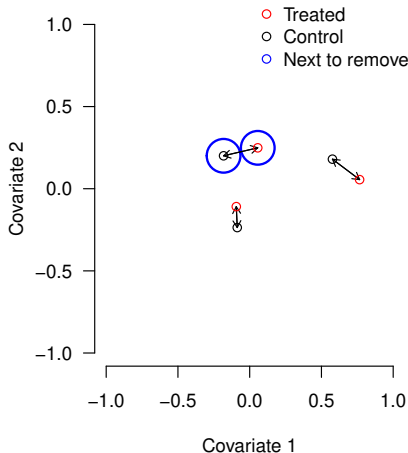


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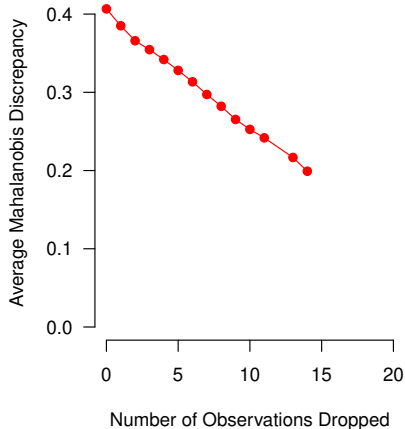


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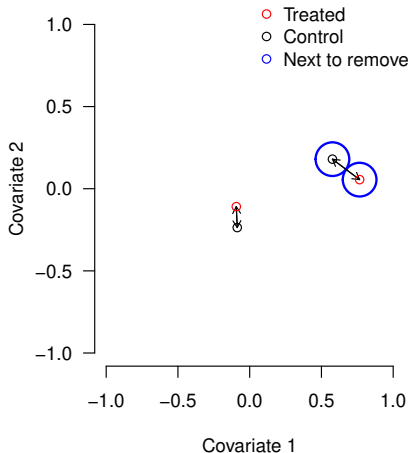


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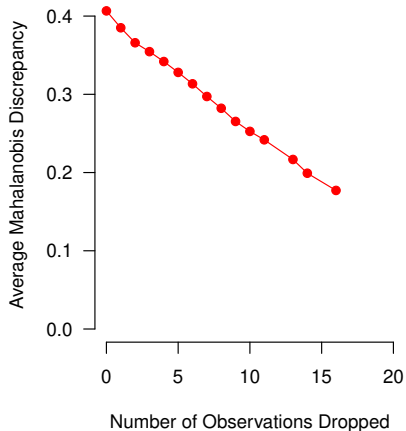


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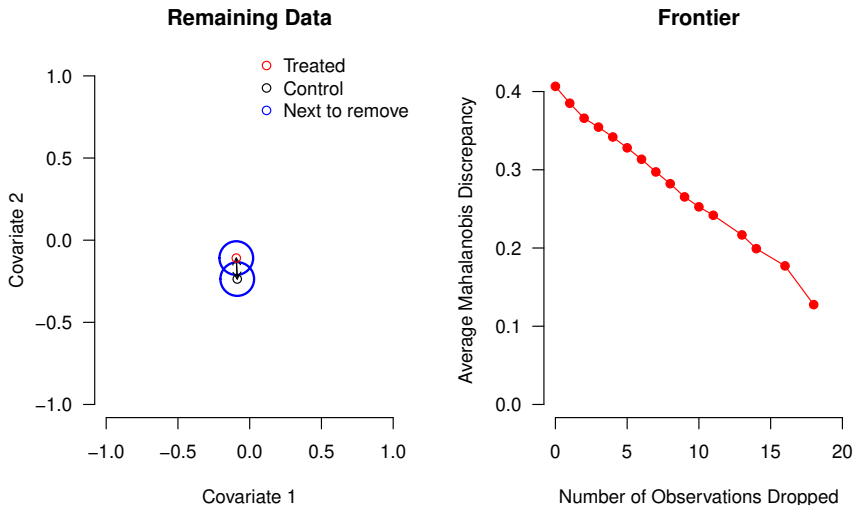
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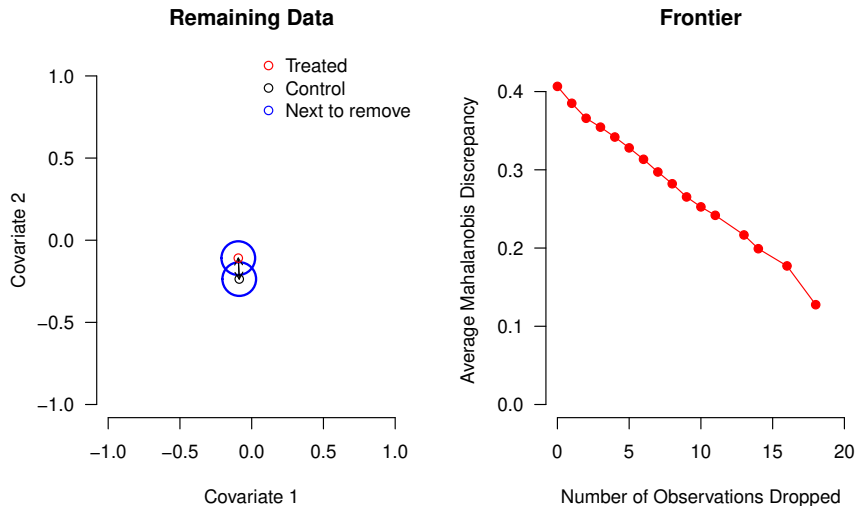
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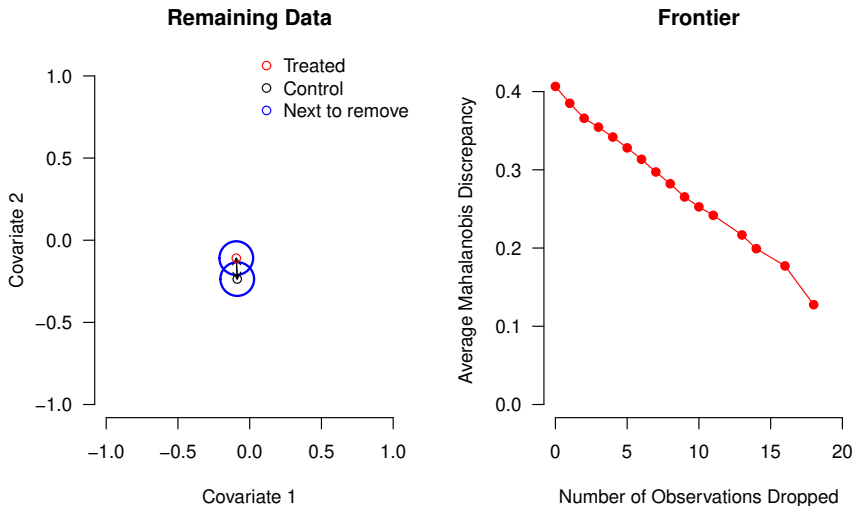


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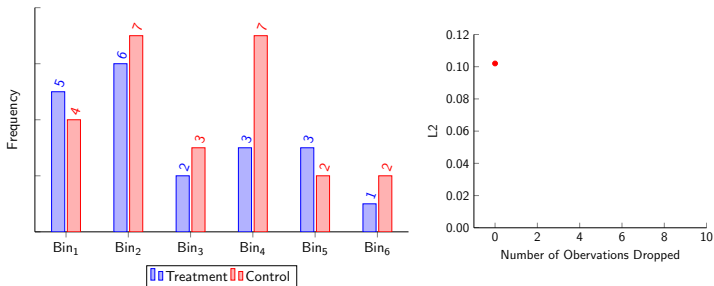
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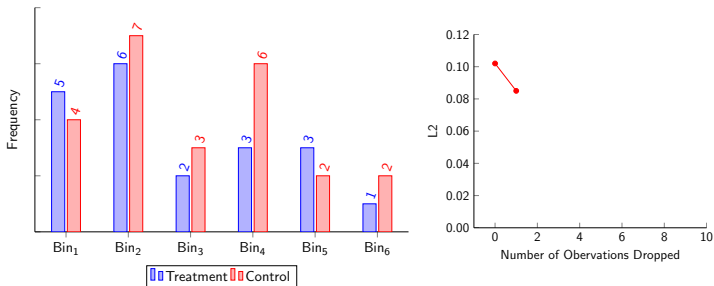


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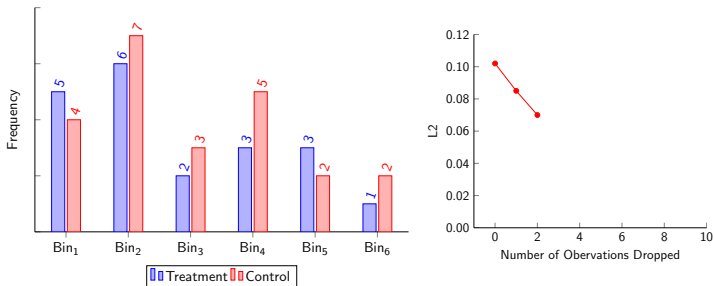
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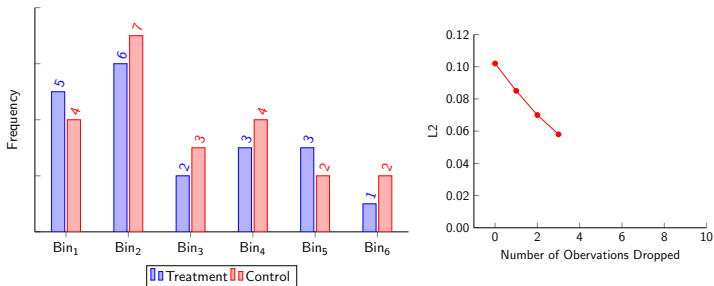
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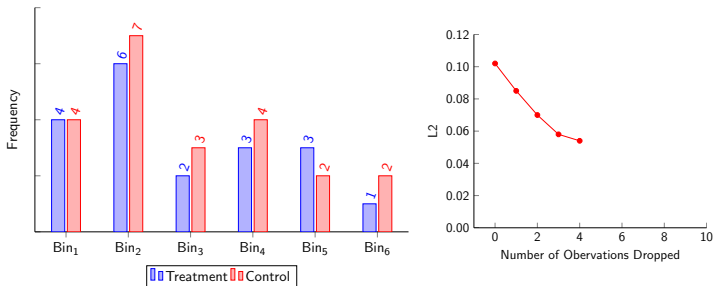
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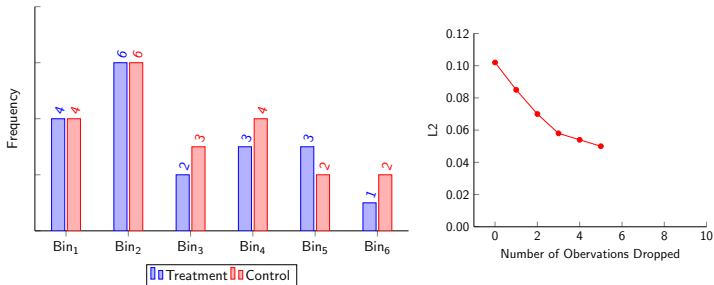
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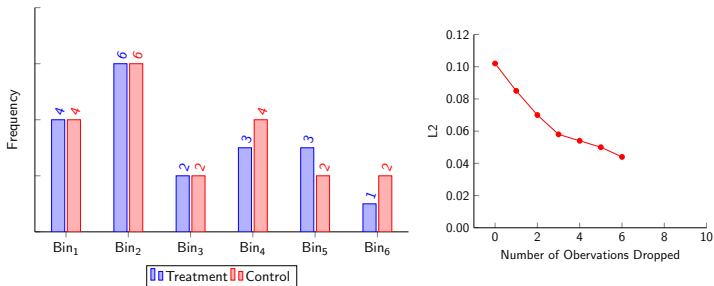
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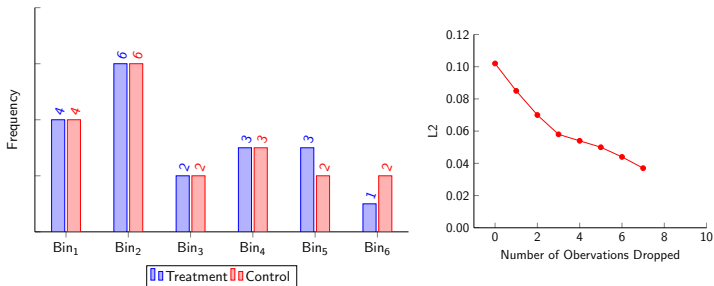
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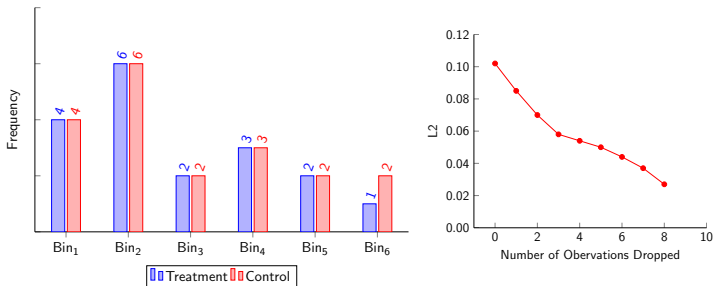
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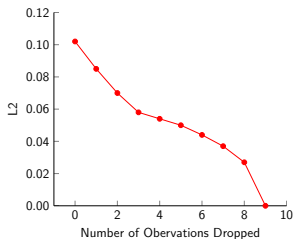
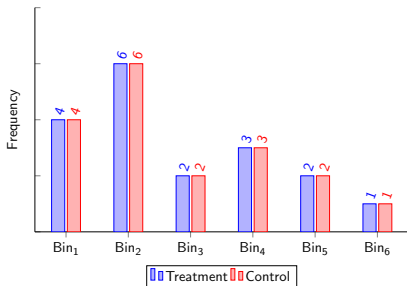
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- **↪ Using more information is simpler and more powerful**

For more information, papers, & software

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